BIO 005 Human Physiology · Week 7 notes · Dr. Sharilyn Rennie

Fuel, fiber types and fatigue

The written notes for the fuel, fiber types and fatigue walkthrough. Week 7 competencies 11 to 15.

To print: these notes as a two-column PDF. Already printed the Week 7 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

Step by step: from fuel to fatigue and recovery

In your textbook: Silverthorn Chapter 13, the parts on muscle energy, fatigue and the classification of muscle fibers.

Where a working fiber spends ATP

1. A motor neuron fires, and an action potential spreads along the fiber's membrane and down the T tubules.
2. Ca2+ is released from the sarcoplasmic reticulum and lets the crossbridges form.
3. Each myosin head splits one ATP on every turn of the crossbridge cycle.
4. The SERCA pump uses ATP to move Ca2+ back into the sarcoplasmic reticulum, so the fiber can relax.
5. The Na+/K+ pump uses ATP to restore the Na+ and K+ gradients after the action potentials.
6. The fiber stores only a few seconds' worth of ATP, so it has to make more as fast as it spends it.

An all-out sprint that becomes a long run, source by source

1. In the first seconds, the ATP already in the fiber is used, and creatine kinase moves a phosphate from creatine phosphate onto ADP to make more ATP in one step. No oxygen is needed.
2. Within roughly 10 to 15 seconds, most of the creatine phosphate is used up.
3. Anaerobic glycolysis carries most of the load next. In the cytosol, glucose is split into two pyruvate, making a net 2 ATP (3 when the glucose comes straight off stored glycogen).
4. Because oxygen cannot be delivered fast enough, pyruvate is turned into lactate, which regenerates NAD+ so glycolysis can continue.
5. After about a minute, glycolysis falls off: H+ builds up and slows its enzymes, and oxidative phosphorylation takes over more of the work.
6. Meanwhile breathing, heart rate and blood flow to the muscle rise, and oxidative phosphorylation in the mitochondria comes up to speed.
7. At a steady pace, oxidative phosphorylation supplies nearly all the ATP, about 30 to 32 ATP in all from each glucose, and can also burn fatty acids. It lasts as long as oxygen and fuel last.

Fatigue during a long steady grip

1. You hold the grip at a steady force, about half your strongest squeeze.
2. The working fibers begin to fatigue, and each one produces less force.
3. To keep the total force steady, the nervous system recruits more motor units and fires them faster.
4. More active fibers make the surface EMG grow larger, even though the force is flat.
5. When no more motor units can be added, the force falls no matter how hard you try.

Oxygen use from rest, through exercise, into recovery

1. At rest, oxygen use is steady at the resting level.
2. When exercise starts, the need for ATP rises at once, but oxygen use rises gradually.
3. Stored ATP, creatine phosphate and anaerobic glycolysis cover the gap. This gap is the oxygen deficit.
4. Oxygen use levels off, and oxidative phosphorylation supplies nearly all the ATP.
5. When exercise stops, oxygen use falls slowly and stays above resting for a while. This is excess post-exercise oxygen consumption (EPOC).
6. The extra oxygen rebuilds ATP and creatine phosphate, reloads myoglobin and hemoglobin, turns lactate back into fuel or glucose, and pays for the higher body temperature and the harder-working heart and breathing muscles.

ATP sources during activity

In your textbook: Silverthorn Chapter 13, the part on how muscle fibers get the ATP for contraction.

Dr. Rennie's videos for this part: How muscle fibers make ATP, the whole picture.

A muscle fiber spends ATP on three jobs: crossbridge cycling, pumping Ca2+ back into the sarcoplasmic reticulum with SERCA, and restoring ion gradients with the Na+/K+ pump. It stores only a few seconds' worth of ATP, so it relies on three ways of making more.

Creatine phosphate (phosphocreatine)

A small molecule in muscle that carries a high-energy phosphate. Creatine kinase transfers the phosphate to ADP, making ATP in one step. Fastest source, no oxygen needed, but lasts only seconds. Refilled at rest using ATP from the mitochondria.

Anaerobic glycolysis

Glucose split into two pyruvate in the cytosol, making 2 ATP per glucose. Pyruvate is turned into lactate, which regenerates NAD+. Fast, no oxygen needed, lasts about a minute of hard work.

Oxidative phosphorylation (aerobic metabolism)

ATP made in the mitochondria using oxygen, about 30 to 32 ATP in all from each glucose. Can also use fatty acids and amino acids. Slowest to start and lowest top rate, but lasts as long as oxygen and fuel last. Supplies nearly all the ATP for resting and moderately active muscle.

Source	How fast it supplies ATP	How long it can keep going	Needs oxygen?
Stored ATP and creatine phosphate	Fastest	Seconds, roughly the first 10 to 15	No
Anaerobic glycolysis	Fast	About a minute of hard work	No
Oxidative phosphorylation	Slowest	Hours, while oxygen and fuel last	Yes

- Rate (the tap) and capacity (the tank) are separate properties, but they run in the same order: the fastest source is the first to run out.
- A 100 meter dash runs mostly on stored ATP and creatine phosphate, with glycolysis coming up quickly. An hour-long run depends almost entirely on oxidative phosphorylation.
- Lactate is not wasted. It leaves the fiber and can be used as fuel by other tissues or turned back into glucose by the liver.

Three kinds of muscle fiber

In your textbook: Silverthorn Chapter 13, the part on the classification of skeletal muscle fibers by contraction speed and resistance to fatigue.

Dr. Rennie's videos for this part: Skeletal muscle fiber types.

Skeletal muscle fibers are classified by how fast they contract and how well they resist fatigue.

Property	Slow oxidative (type I)	Fast oxidative-glycolytic (type IIA)	Fast glycolytic (type IIX)
Speed of contraction	Slow	Fast	Fast
Fiber diameter	Small	Medium	Large
Color	Dark red	Red	Pale (white)
Myoglobin	High	High	Low
Mitochondria	Many	Many, fewer than slow oxidative	Few
Capillaries	Many	Many	Few
Main ATP source	Oxidative phosphorylation	Oxidative and glycolytic	Anaerobic glycolysis
Fatigue resistance	Resists fatigue	Intermediate	Tires quickly
Typical task	Posture, held for hours	Walking, moderate sustained work	Jumping, sprinting, lifting a heavy load

Myoglobin

A red protein in muscle that binds oxygen and helps move it into the fiber. It makes oxidative fibers red.

Fast-twitch fibers

Fast oxidative-glycolytic and fast glycolytic fibers. Their myosin splits ATP faster, and they pump Ca2+ back into the sarcoplasmic reticulum faster, so their twitches rise and fall sooner.

Slow-twitch fibers

Slow oxidative fibers. Slower myosin and slower Ca2+ uptake, so a longer twitch.

- A small diameter keeps the distance oxygen must travel from the capillaries short, which helps a slow oxidative fiber rely on oxidative phosphorylation.
- Every muscle is a mix of fiber types. All the fibers in one motor unit are the same type.
- The proportions differ between muscles and between people, set partly by genetics and partly by use.
- Small motor units of slow oxidative fibers are recruited first for light work. The large motor units of fast glycolytic fibers are added only when a task needs a lot of force.

Fatigue

In your textbook: Silverthorn Chapter 13, the part on muscle fatigue.

Dr. Rennie's videos for this part: Muscle fatigue, Types of fatigue and what causes them.

Muscle fatigue is a drop in the force a muscle can produce or keep up despite continued effort. It is reversed by rest.

Central fatigue

Fatigue that starts in the central nervous system, the brain and spinal cord. It includes feeling tired and wanting to stop, and it often sets in before the muscles themselves fail, which may help protect the body.

Peripheral fatigue

Fatigue that starts anywhere from the neuromuscular junction to the crossbridges: at the junction, in the action potential along the T tubules, in Ca2+ release from the sarcoplasmic reticulum, or at the crossbridges.

Contributors to peripheral fatigue

- K+ leaves the fiber with each action potential and builds up in the T tubules, changing the membrane potential and reducing Ca2+ release.
- Inorganic phosphate (Pi) builds up as ATP and creatine phosphate are broken down. It interferes with the power stroke and with Ca2+ release, and it is thought to be one of the main causes of fatigue in short bursts of maximal effort.
- Less Ca2+ is released from the sarcoplasmic reticulum, so fewer crossbridges form.
- In long, moderate exercise, fatigue goes along with using up the fiber's glycogen.
- Which factor matters most depends on the kind and intensity of the exercise.
- ATP does not run out completely: the fiber slows down before its ATP is gone.

- Lactic acid (lactate and H+) is no longer thought to be the main cause of muscle fatigue. The fall in pH may still play some part, and research on it continues.

Grip force and EMG (Competency 15, done in lab)

- A grip dynamometer records grip force. Surface EMG electrodes on the forearm record the combined electrical activity of the muscle fibers underneath.
- During a steady submaximal hold, force stays flat while EMG amplitude grows, because more motor units are recruited and fired faster to replace fatiguing ones.
- When recruitment can rise no further, force falls.
- To tell central from peripheral fatigue, the motor nerve can be stimulated electrically once a person can no longer squeeze hard. If force comes back, the failure was central. If force stays low, the fatigue is in the muscle.

Oxygen use after exercise

In your textbook: Silverthorn Chapter 13, the part on muscle fatigue and oxygen, and Chapter 25, the part on oxygen consumption during and after exercise.

Oxygen deficit

The gap at the start of exercise between the oxygen the work needs and the oxygen actually used. Stored ATP, creatine phosphate and anaerobic glycolysis cover it while breathing, heart rate and blood flow rise.

Excess post-exercise oxygen consumption (EPOC)

Oxygen use that stays above resting after exercise ends. It used to be called oxygen debt. It is usually larger than the oxygen deficit, because it pays for more than replacing the deficit.

What the extra oxygen after exercise pays for

- Rebuilding ATP and creatine phosphate through oxidative phosphorylation.
- Reloading myoglobin in the muscles and hemoglobin in the blood with oxygen.
- Turning lactate back into pyruvate to use as fuel, or back into glucose in the liver.
- A higher body temperature, which speeds up metabolism.
- The heart and breathing muscles, which are still working harder than at rest.
- After a sprint, recovery leans more on rebuilding creatine phosphate and clearing lactate.

Training and disuse

In your textbook: Silverthorn Chapter 13, the parts on muscle fiber types and on how exercise changes muscle.

Dr. Rennie's videos for this part: Adapting to training.

Muscle fibers change with how they are used. This is called plasticity.

Change	Fiber size	Myofibrils	Mitochondria	Capillaries
Months of endurance training	Little change	Little change	Many more	More
Months of resistance training	Much larger (hypertrophy)	Many more	Do not keep pace, spread over a larger area	Do not keep pace, spread over a larger area
Six weeks in a cast (disuse)	Smaller (atrophy)	Fewer	Fewer	Fewer

Hypertrophy

An increase in fiber size, mainly from building more myofibrils with more actin and myosin. The number of fibers changes little.

Disuse atrophy

Shrinking of fibers that are hardly used: more protein is broken down than built. The muscle becomes weaker and tires sooner. It can be rebuilt with use as long as the nerve supply is intact.

- Endurance training also raises myoglobin, and some fast glycolytic fibers become more oxidative, behaving more like fast oxidative-glycolytic fibers.
- Strength depends mostly on how many myofibrils lie side by side. Endurance depends on mitochondria and blood supply. A resistance-trained fiber is the stronger one, and an endurance-trained fiber keeps going longer.
- What this means for you: people on bed rest or in a cast lose muscle quickly, which is one reason early movement and physical therapy are part of recovery. Astronauts in low gravity face the same problem.